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BCIS R&D Project

Reflective Report Requirements
and Assessment Guide

Date	Version	Author	Notes
17 April 2014	1.0	Anne Philpott	Contains collated notes from earlier project assessments resources.
February 2015	1.1	Anne Philpott	Minor updates
July 2015	1.2	Anne Philpott	Minor updates
May 2016	1.3	Stephen Thorpe	Template revision, correction and addition of missing reference and minor updates
May 2017	1.4	Stephen Thorpe	Minor updates
Aug 2020	1.5	Sarita Pais	Assessment weightage

1. Introduction

The formal reflective report offers an opportunity for students to reflect upon the significance of their work and what they may have gained personally and professionally from the R&D experience as well as identifying opportunities for further development. The report should include a critical dimension and be related to relevant literature, such as that by Argyris & Schön (1974; 1996) and Schön (1987) discussing reflective practice, the nature of professionalism and the concept of theories of action.

Fincher, Petre and Clarke (2001) in their book entitled *Computer Science Project Work: Principles and Pragmatics* place special emphasis upon the value of reflection: “*Reflection on experience underpins the process of successful learning and is essential to the success of education.*” (p. 226). Engaging in reflective practice creates a mind-set that is invaluable for workplace professional performance well beyond your degree (read it’s a skill-set for attaining future success and promotion).

The Fincher et al. model is drawn from earlier work by Argyris and Schön (1974; 1996) and Schön (1987) in which professional work involves self-monitoring, continual improvement and cycles of planning, putting things into action, observing how they work, and then reflecting on them for further rounds of plan, act, observe, reflect.

The learning goals associated with the assessment are:

1. *Show the ability to successfully undertake original work and demonstrate a professional attitude.*
2. *Demonstrate the ability to integrate the different skills required to bring a project to a successful conclusion.*
3. *Communicate effectively with clients and sponsors in both written and verbal presentations and in group situations.*
4. *Effectively manage, monitor and control the activities involved in a research and development project.*
5. *Determine an appropriate process and accompanying set of deliverables for their project by selecting and justifying an appropriate methodology for their project.*
6. *Show the ability to document appropriately the deliverables for their project (e.g. software specifications, project plans, source code, technical reports, white papers, literature reviews, academic articles for publication etc.), as agreed with the project supervisor(s).*

2. Reflection throughout the project

Reflective work in your project is aimed at developing your capabilities through:

- Before the project students are required to consider their professional and personal needs, and in the project proposal/learning agreement submission indicate the skills and knowledge involved in order for them to successfully complete their projects.
- Students maintain a personal journal or work log book, to track activities and time spent on the project and to note events and your own reflections upon them as they occur.

- At the completion of the project the final poster presentation offers students an opportunity to reflect upon and demonstrate what they may have learnt, the results they have collectively achieved, and the problems they have successfully overcome.
- The Reflective Report offers an opportunity for students to reflect upon the significance of their work, and what they have gained personally and professionally from the experience and where they may have opportunities to develop.

A reflective practitioner model often surfaces personally embarrassing and uncomfortable issues, aspects of ourselves we may not want to explore - especially for those unpractised or intolerant to constructive criticism. There may be many flawed assumptions throughout your project that must be critically addressed to avoid costly errors (or even disastrous project failure) and the genius of theory often starts in a practice situation.

In terms of what's expected consider an apple falling on Neuton's head in his world of scientific practice leading to a profound understanding of a fundamental law of the universe - the Universal Law of Gravitation. So too will be moments throughout your project that contribute some brand new learnings.

Now Neuton didn't complain about his sore head or the location of the tree, or even care too much that it was an apple and not a plum falling on his head – he was interested in knowing a much deeper why? What force, or forces were at play in order for made apple to move from there to here.



Newton's apple tree at Woolsthorpe

For students contributing honestly to a journal or log book consistently and thoughtfully maintained throughout the project, and well considered self-evaluation in the reflective report can prove effective tools in this project's journey. What were the forces at play around the critical incidents that had the largest impact on your project? On the people involved? On the client? What might be observed? What could you have done differently to create a different outcome?

3. Content of the Reflective Report

The focus of the report **must not** be a description of your project activities that you have undertaken, but focus on critical analysis, evaluation and reflection **on how you performed**, and your ability to demonstrate the application of relevant theory to the project experience. Theory may relate to processes, technology, or frameworks to which you have been exposed in prior courses, or encountered in practice and have taken the time to read further and reflect upon. For example, project management processes, group dynamics, negotiation, modelling, design patterns, intellectual property law, data coupling and cohesion, pair programming are all examples of areas in which relevant theory could be compared to your project experiences.

Actively link specific examples of situations that you encountered in your project to the frameworks, processes, techniques, practices or guidelines outlined in the literature. Credible academic sources should be used, books, journals and conference articles (such as those found in the ACM or IEEE Explore digital libraries - not websites or Wikipedia).

These could be in the form of Critical Incidents (see Fook and Cooper, 2003) as discussed in the class on Addressing Conflict, which includes a structure for articulating your reflections

and learning for

Feedback received from your client can be extremely helpful in preparing a reflective report. It is therefore strongly recommended that you provide the feedback form to your client to fill in before you write your reflective report. This form will be returned directly to your supervisor, who will in turn give it back to you so you will need to plan this step in advance, at least two to three weeks before the scheduled submission date for your report, to allow you time to consider and reflect on the feedback.

The reflective report should illustrate an ability to reflect upon your experiences and show awareness of your own personal and professional growth as a result of the project. One implicit assessment aspect is that all work be produced to a level of quality that would be acceptable in a professionally managed commercial environment. You should therefore ensure the quality of the presentation of your report by making use of tools that will help (for example, spelling and grammar checking), and by proof reading the report carefully. The report format and some guidelines for content for the reflective report are outlined in Appendix B below.

4. Submission and Marking Process

The reflective report contributes 10% to your final mark. Each student prepares and submits their own report. It is due at the end of week 12 of your final semester in the project and at the same time as your Project Portfolio. Specific deadlines will be notified using Blackboard.

You must submit your report using the Turnitin link provided on Blackboard. Additionally two hard copies of the report need to be submitted (one is needed for your supervisor and one for the moderator). Appendix A provides the marking guideline that will be used to mark your reflective report.

References

- Argyris, C. (1996). Unrecognized defenses of scholars: Impact on theory and research. *Organization Science*, 7(1), 79-87.
- Argyris, C. & Schön, D. (1974). *Theory in practice: Increasing professional effectiveness*. San Francisco, CA: Jossey Bass.
- Fincher S., Petre M. & Clark M. (2001). *Computer science project work: Principles and pragmatics*. London: Springer.
- Fook, J., & Cooper, L., cited in (2003) *Bachelor of Social Work Fieldwork Manual*, Dept. Social Work, School of Primary Health Care, Monash University.
- Hinchcliff, J. (1974). *Values Integrating Education: An Exploration of Learning in New Zealand*. Pukekohe, NZ: Mirilea Press.
- Schön, D. (1987). *Educating the Reflective Practitioner*. San Francisco, CA: Jossey Bass.

CRITERIA	FEEDBACK
Cover page (must have Title, Project, Name, Student ID, Date, and Supervisor's Name). Abstract ½ page Acknowledgments ½ page Table of contents 1 page Introduction 2-3 pages	
Project Evaluation (2-3 pages) - Evaluation of the project achievements against those outlined in the original project proposal/learning agreement - Evaluation of the significance of the project to the organization (including both the development of the project in the short term and the long term significance)	
Link between theory and practice (2-3 pages) - Analysis of the project with regard to relevant theory (such as frameworks, models, processes), including critical evaluation of the application of those theories to the project - Critique of the research or development methodology adopted in the conduct of the project	
Personal and professional development (2-3 pages) - Evaluation of own personal and professional growth against the originally expected skills and knowledge to be acquired in the project proposal/learning agreement. - Identification of future learning needs. Evaluation of the team roles, the effectiveness of the team members, and own	

interaction with team members if the project was a group project. For individual projects evaluation of the limitations of working alone, the nature of individual processes, and their impact upon your project.	
Summary and conclusion of total project experience (1- 2 pages)	
References Content should be linked to relevant literature and references are expected to be from quality sources, and correctly cited in the body of the report	
Index of appendices Appendices	

Reflective Report Grade

A+	A	A-	B+	B	B-	C+	C	C-	D	D
									Over 40%	Under 40%

Signed by: _____
(Project Supervisor)

Date _____

Signed by: _____
(Moderator)

Appendix B - Reflective Report Content Guide

A suggested format:

Font size 12, 1.5 spaced, no more than two standard fonts.

Cover page with a Title, Project, Name, Student ID, Date, and Supervisor's Name.

Abstract	½ page
Acknowledgments	½ page
Table of contents	1 page
Introduction	2 - 3 pages

Project evaluation (2-3 pages)

- Evaluation of the project achievements against those outlined in the original project proposal/learning agreement. *What do you plan to do? What did you actually do? Did this achieve the goals?*
- Evaluation of the significance of the project to the organisation (including both the development of the project in the short term and the long term significance). *Why does it matter to them?*

Link between theory and practice (2-3 pages)

- Analysis of the project with regard to relevant theory (such as frameworks, models, processes), including critical evaluation of the application of those theories to the project. *Areas of theory that are relevant could include such things as collaboration, project management, technology related theory, tools processes and practices etc. Review the theory you have learnt in earlier papers in your degree and identify how your project experience supported or did not support the theories.*
- Critique of the research or development methodology adopted in the conduct of the project. *What worked? What did not? Why? IS this what the literature would predict?*

Personal and professional development (2-3 pages)

- Evaluation of your own personal and professional growth against the originally expected skills and knowledge to be acquired in the project proposal/learning agreement. *Have you actually learnt what you expected to learn? Are there other things you have learned that you did not expect?*
- Identification of future learning needs. *What do you think your current gaps and weaknesses are? How could address them*
- Evaluation of the team roles, the effectiveness of the team members, and your own interaction with team members if the project was a group project. For individual projects, evaluation of the limitations of working alone, the nature of individual processes, and their impact upon your project.

Summary and conclusion of total project experience (1- 2 pages)

References

- A suitably comprehensive set of references from quality sources, **for works correctly cited in the body of the report**. Note that references must be **APA 7th format**. *How many is enough? Well obviously quality and relevance counts so more irrelevant or low quality references would not be better than fewer good relevant ones. However we would definitely expect there to be more than five and probably more than ten.*

Index of appendices.

If there is something that you believe is relevant and belongs in an appendix.

Appendix(ices)

No formal limit on size or content

Please hand in **two** printed copies for assessment. These will be treated as personal confidential reports and will not be shown to others unless for marking and moderation purposes.

Note: Past students who have linked **specific examples** of issues or experiences to academic journal articles, books or refereed conference papers presented higher **critical thought** and received significantly higher grades.

The aim of this assessment is to offer you an opportunity to demonstrate your insights about what you have learnt or what you have discovered through direct experience.

Consider questions such as: Is the theoretical framework, model, process, strategy, practice or technique we have followed realistic in real practice? Does it have or weaknesses deficiencies? Do you have any recommendations related to the theories you have read? What have I really learnt about myself in following this approach?